Chapter 39
Market Risk Capital and Model Risk

Chapter 38 built measures of what a book can lose. This chapter is about what a regulator does with them, and about the exposure that no measure in Chapter 38 can see: the risk that the model producing the numbers is wrong.

The two subjects belong together for a reason that becomes obvious once stated. A capital rule is a contract with a bank about a number, and the number comes from a model the bank builds. So the rule must either specify the model or test it, and the modern framework does both—which is why so much of it is machinery for deciding whether a bank's own model may be used at all.

One caution comes before any of the detail. The implementation dates in this area have moved repeatedly and in every major jurisdiction. As of November 2026, they stood as follows:

- the Basel standard's own implementation date was January 1, 2023;
- Canada and Japan had applied it since late 2023 and March 2024 respectively;
- China had applied it since January 1, 2024;
- the European Union was due to apply its market risk rules from January 1, 2027, after two postponements, with a temporary multiplier, adopted in June 2026, offsetting the capital increase until the end of 2029;
- the United Kingdom was due to apply them from the same date, with its internal models approach for market risk deferred to January 1, 2028; the Prudential Regulation Authority (PRA) consulted on adjustments to it in June 2026;
- the United States had re-proposed its version in March 2026 with no final rule.

Check the calendar against the date you read this. The structure below is stable, and the calendar is not.

Sections 39.1 and 39.2 explain why the trading book has its own capital rule and where its boundary sits. Sections 39.3 to 39.8 are the rule itself: the standardized approach, internal models, non-modellable risk factors, the P&L attribution test, desk-level backtesting, and the approval cliff those tests create. Sections 39.9 to 39.12 assemble the total and cover the three channels that sit above or beside it: the output floor, prudent valuation, and stress testing. Section 39.13 asks what the framework changed and who bears the cost, and Sections 39.14 and 39.15 turn to model risk, including the validation of models that cannot be verified.

39.1 WHY THE TRADING BOOK NEEDS ITS OWN RULE

A bank keeps two books. The loans and bonds in the banking book are mostly held to maturity and capitalized against default. The positions in the trading book are held to be sold, so the loss that matters is a price move over a short horizon, not a default over years.

The framework that preceded the current one was built on Chapter 38's VaR and failed in 2008 in three identifiable ways. It used a ten-day 99% VaR calibrated on recent data, so it was

procyclical exactly as Section 38.18 describes. It capitalized the trading book far more lightly than the banking book for the same economic exposure, which produced a decade of moving assets across the boundary. And it assumed every position could be exited in ten days.

The current framework is the Basel Committee's **Fundamental Review of the Trading Book**, known by its initials, **FRTB**, which is the name to search for; the US implementation was folded into the package called the **Basel III endgame**, whose scope has been contested and revised since it was first proposed. It answers each of the three failures:

- *procyclicality*, by calibrating the internal-model measure to a period of stress rather than to recent data, so that it cannot fall because markets are calm (Section 39.4);
- *the light trading book*, by fixing which book a position belongs in with a prescriptive list, and by denying any capital benefit to moving it (Section 39.2);
- *liquidity*, by giving each risk factor a horizon of 10 to 120 days, so that a risk factor that takes longer to exit is charged over a longer horizon (Section 39.4).

It is worth reading that way: every complication below exists because something specific broke. The framework offers two ways to compute the charge, the **standardized approach**, which every bank computes, and the **internal models approach**, which a bank may use desk by desk once its models are approved.

Case Study 39.1 The patch, and why it was not enough

The response to 2008 came in two stages, and the gap between them is instructive.

The immediate patch arrived in July 2009 and is known as **Basel 2.5**. It added a stressed VaR computed on a crisis window, and an **incremental risk charge** for default and migration in the trading book. Both were added to the existing measure rather than replacing it, and the Basel Committee's (2009) impact study put the combined effect at two to three times existing trading-book capital.

That the patch worked is not in dispute; that it was a patch is the point. Stressed VaR addressed procyclicality by adding a second calculation rather than fixing the first, so banks ran two measures on the same book and held the sum. The incremental risk charge covered default in the trading book because the boundary had let credit exposures migrate there, treating the symptom rather than the boundary; it did use liquidity horizons, with a three-month floor, but only for default and migration. Nothing in the patch addressed liquidity in the VaR measures themselves: a position was still assumed exitable in ten days.

Of the three failures above, the patch had met only procyclicality, and only by adding a calculation. The boundary and the horizon problems survived it, and so did a defect in the measure itself; those three were left to the current framework. The boundary was still intent-based, until Section 39.2's prescriptive list. The horizon was still a constant ten days, until Example 39.3's schedule of liquidity horizons. And the measure was still VaR, with Section 38.4's subadditivity failure and Section 38.8's finding that a fatter tail can lower VaR intact. That is the direct reason the current framework is written around expected shortfall.

The general lesson is about how frameworks fail. The 2009 patch was fast, effective in raising capital, and left every structural defect in place. Its successor took a decade and is still being implemented: a fast patch or a slow cure is the choice a supervisor faces whenever the diagnosis is architectural and the pressure is immediate.

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39.2 THE BOUNDARY BETWEEN THE TRADING BOOK AND THE BANKING BOOK

The first fix concerns the line between the two books of Section 39.1: the trading book, capitalized against price moves under this chapter's rules, and the banking book, capitalized against default under the credit rules of Section 27.11 and Section 39.10. It is the least mathematical fix and was the most consequential. If capital differs between the two books, the boundary between them is a valuable option, and banks exercised it. A bank holding $500 million of corporate bonds that attracted 3% capital in the trading book and 8% in the banking book had a five-point capital difference riding on a classification, and under the old rule the classification turned on the bank's stated intent.

The current rule makes the boundary prescriptive rather than intent-based. A presumptive list says what sits where:

- instruments held for short-term resale, positions in the correlation trading portfolio (securitization tranches and *n*th-to-default credit derivatives), net short credit or equity positions, and anything arising from underwriting or market making are presumed to be in the trading book;
- unlisted equities, real estate, retail and small-business credit, and equity investments in funds whose holdings the bank cannot identify and value daily are in the banking book.

A bank may depart from the presumption only with supervisory approval and documented reasons. Reassignment between the books requires supervisory approval, is permitted only in extraordinary circumstances, and may not produce a capital benefit: if moving a position lowers its charge, the bank holds a disclosed surcharge equal to the difference. The option was not tightened; it was removed, which is the right response to an incentive rather than a modeling problem.

39.3 THE STANDARDIZED APPROACH

Every bank must compute the standardized charge, whether or not it has model approval, and the charge has three parts.

The sensitivities-based method is the bulk of it. A desk reports delta, vega, and curvature sensitivities to prescribed risk factors; the regulator supplies risk weights and correlations. The charge then combines them by a prescribed formula: each sensitivity is multiplied by its risk weight, and the products are combined the way a portfolio's standard deviation is built from the variance of equation (38.4) in Chapter 38's model-building approach, as the square root of their squares plus their correlated cross-products. Example 39.1 works it through. The three sensitivities are the coefficients of Chapter 18's expansion, equation (18.1), read as a capital rule: delta charges its first-order term, vega its volatility term, and curvature its second-order term. The correlations are given, not estimated. Section 38.11 showed that correlation is the least stable input to a risk measure: estimated in calm periods and applied in violent ones, so that a long book's measured risk comes out too low. The framework settles the question by fiat: the supervisor would rather set the number and charge for doubt about it than depend on a bank's estimate.

The default risk charge covers **jump-to-default**, the loss if an issuer defaults overnight with no warning in its spread, on traded bonds, credit derivatives and equities. **The residual risk add-on** is a flat charge on payoffs the sensitivities do not describe. The next three subsections take delta and vega, curvature, and the other two parts.

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39.3.1 Delta and vega

Sensitivities are grouped into **buckets** (one currency's interest rate curve, one equity sector) and aggregated in two stages, within each bucket and then across buckets. The delta charge is worth working through, because its structure contains a device that has no analog in Chapter 38 and that determines which books the framework treats harshly.

Example 39.1 Three correlation scenarios, and which one binds

Consider one currency and three points on its interest rate curve, at 2, 5 and 10 years. For each point k the bank reports its net sensitivity s_k, the loss on all its positions on this curve when the rate at that point alone rises by one basis point. This is Section 7.1's bucketed DV01, in dollars per basis point, so receiving and paying fixed at the same point carry opposite signs. The regulator supplies a risk weight RW_k for each point, here 130, 110 and 110 basis points (not the percentage of an exposure that Section 27.11's credit weights are, but the size of the rate move it prescribes as a shock), and the correlations

$$\rho_{kl} = \max\left(e^{-0.03\,|T_k - T_l|/\min(T_k, T_l)},\ 0.40\right),$$

where T_k is the tenor of point k in years. These give 0.9560 between 2 and 5 years, 0.9704 between 5 and 10 years, and 0.8869 between 2 and 10 years. Both the risk weights and this correlation are the Basel standard's own, not choices made for the example (for the major currencies a bank may divide the weights by $\sqrt{2}$). The correlation belongs to the family Chapter 32 used for forward rates, Example 32.1's exponential decay and the floored blend that follows it, with two differences: the distance is measured relative to the shorter tenor, so the short end decorrelates faster (2 and 5 years at 0.956, against 0.991 for the same three-year gap between 10 and 13 years), and the floor of 0.40 is a cutoff rather than a blend. The **weighted sensitivity** $WS_k = s_k \times RW_k$ is then the dollar loss from that prescribed move at point k, and the bucket charge is

$$K_b = \sqrt{\max\left(0,\ \sum_k WS_k^2 + \sum_{k \neq l} \rho_{kl}\, WS_k\, WS_l\right)},$$

where the floor at zero matters only when the correlations are moved, as they are next.

The charge is computed three times, once with the prescribed correlations and once with each of

$$\rho_{kl}^{\text{high}} = \min\left(1.25\,\rho_{kl},\ 1\right), \qquad \rho_{kl}^{\text{low}} = \max\left(2\rho_{kl} - 1,\ 0.75\,\rho_{kl}\right).$$

The capital number is the largest of the three charges, not the charge at the smallest correlations: which set of correlations binds depends on the book, as the two books below show. (Basel takes this maximum after the buckets have been aggregated; with a single bucket, as here, that is the same thing.)

Take two books. The first receives fixed at all three points, with sensitivities of \$2,000, \$3,000 and \$4,000 per basis point. The second is a butterfly, the kind of curve trade Example 38.11 measures, here with wings at 2 and 10 years and the body at 5: it receives fixed at 2 and 10 years (\$2,200 and \$2,000 per basis point) and pays fixed at 5 years (\$4,600), sized so that its weighted sensitivities sum to zero: the prescribed shocks, 130 basis points at 2 years and 110 at 5 and 10, leave it unchanged, although a parallel one-basis-point move changes its value by \$400.

For the directional book the weighted sensitivities are \$260,000, \$330,000 and \$440,000, so under the prescribed correlations

$$\begin{aligned} K_b^2 = {} & 260^2 + 330^2 + 440^2 + 2 \times (0.9560 \times 260 \times 330 \\ & + 0.8869 \times 260 \times 440 + 0.9704 \times 330 \times 440) = 1{,}018{,}894, \end{aligned}$$

in thousands of dollars squared and with the correlations unrounded, so K_b = \$1,009,403. The three scenarios give:

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Scenario	Directional book	Butterfly
Low correlation	\$988,376	**\$100,854**
Medium (as prescribed)	\$1,009,403	\$71,315
High correlation	**\$1,030,000**	\$0
Charge (worst)	**\$1,030,000**	**\$100,854**
Excess over medium	+2.0%	+41.4%

The two books are punished by opposite scenarios. The directional book gets worse as the curve moves together, so *high* correlation binds, and the penalty is mild, 2.0%, because a directional book is nearly as risky under any correlation. The butterfly is the reverse: under high correlation its charge falls to *exactly zero*, since a perfectly correlated curve makes the hedge perfect, while under low correlation the legs stop offsetting and the charge is 41.4% above the prescribed case.

So the hedged book pays the larger proportional penalty, and the reason is the design working as intended: the rule refuses to let a bank choose the correlation that flatters its book. A framework that used a single prescribed correlation could be arbitraged by constructing positions that are riskless under it: a book hedged only at the prescribed number is unhedged at any other. Taking the worst of three removes that possibility.

The cost is borne by genuine relative-value business. A butterfly is a real hedge whose risk really is small when the curve moves together, and the framework charges it as though the correlations it relies on might not hold. Section 38.11 showed that correlations fail in exactly the episodes capital exists for, so the charge is defensible even though it is expensive.

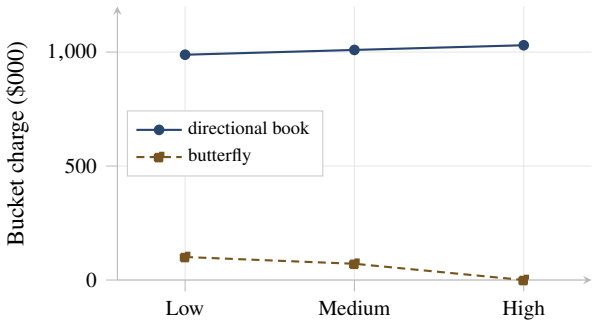


Figure 39.1: Bucket charge under the three correlation scenarios, for the two books of Example 39.1.

Figure 39.1 shows why one rule serves both books. The two lines slope in opposite directions, so no single correlation is conservative for both—and any rule that picked one would leave the other exploitable. Evaluating three scenarios and taking the worst is a simple way to be conservative against a quantity no one can estimate reliably.

Vega is aggregated by the same two-stage formula, with correlations of its own ([TN39] works a book of options through it). Its risk weight is not a fixed number but $\min\big(55\% \times \sqrt{\mathrm{LH}/10},\ 100\%\big)$, where LH is the liquidity horizon the framework assigns to the volatility, in days: 20 days for the volatility of a large-cap stock, giving 77.8%, and 40 days or more for every other volatility, giving 100%.

39.3.2 Curvature

Curvature is charged separately because delta and vega miss the convexity that Chapters 34 and 35 showed dominates structured books. The charge, CVR, shocks the factor up and down by its delta risk weight, revalues in full, and asks how much worse the result is than the delta

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prediction:

$$\text{CVR} = \max\Big(0,\ -\min_{\pm}\big[V\big(S(1 \pm RW)\big) - V(S) \mp RW \times \Delta_{\$}\big]\Big).$$

Here $V(S)$ is the position's value at the stock price S, $\Delta_{\$}$ its dollar delta as in Chapter 38, and RW the delta risk weight. It revalues rather than multiplying a reported gamma by $\frac{1}{2}(\Delta S)^2$ because the shocks are large, and Section 18.10 showed that the expansion's gamma term misstates a large move. The charge has an asymmetry that surprises people: convexity decides its sign, and the floor at zero turns that sign into all or nothing. (Basel applies that floor when it aggregates the risk factors; for a single factor, as here, the effect is the same.)

Take a long \$10 million notional at-the-money six-month call on a large-cap emerging-market stock, at 25% volatility with interest rates at zero. At the money $d_1 = 0.25\sqrt{0.5}/2 = 0.0884$, so the option is worth $\$10\text{m} \times [N(0.0884) - N(-0.0884)] = \$704{,}000$ and its delta is $N(0.0884) = 0.5352$, a dollar delta of \$5.352m. The stock's delta risk weight, and so its curvature shock, is 55%, a Basel weight for a large-cap emerging-market stock (45% to 60% by sector there, 30% to 50% in advanced economies). Up 55%, the option is worth \$5.505m against a delta prediction of $\$0.704\text{m} + 0.55 \times \$5.352\text{m} = \$3.648\text{m}$; down 55%, it is worth almost nothing against a prediction of −\$2.239m. Revaluation beats the delta prediction by \$1.86m on the upside and \$2.24m on the downside, both favorable, because the holder is long gamma. The bracketed term is positive in both directions, so its minimum is positive, and the floor sets the charge to zero. Sell the same option and both figures change sign, giving a charge of **\$2.24m**. Figure 39.2 draws the two gaps.

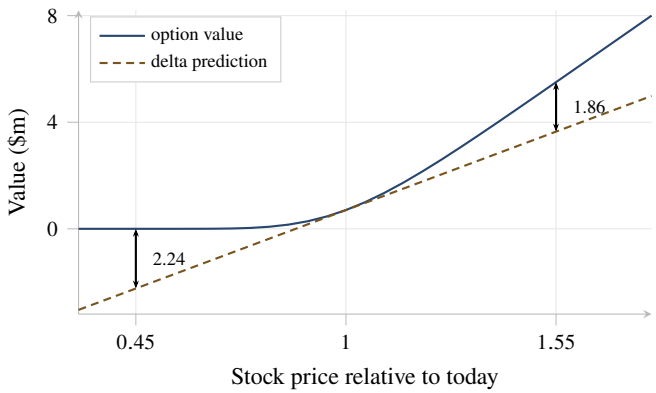


Figure 39.2: A long call's value and its delta prediction, with the gaps at the ±55% curvature shocks. Illustrative.

Both gaps lie on the same side of the line because the value curve is convex; the floor then removes the long holder's negative charge. Long optionality attracts no curvature charge and short optionality attracts all of it, which is the correct treatment and worth stating plainly because it is the first place in this book where a capital rule agrees exactly with Chapter 18's economics. The vega charge completes the picture, and the three components together answer a question a trading desk will ask immediately: does hedging reduce the charge?

Example 39.2 A delta-hedged short option position

Sell the \$10 million notional six-month at-the-money call above, on the same large-cap emerging-market stock at 25% volatility, and delta-hedge it.

Vega is reported as the sensitivity to a *relative* move in implied volatility, the desk's vega multiplied by the volatility itself, so the option's vega of $\$10\text{m} \times \sqrt{0.5}\, N'(0.0884)/100 = \$28{,}099.50$

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per volatility point is reported as $28{,}099.50 \times 25 = \$702{,}488$. The volatility of a large-cap stock has a 20-day liquidity horizon, so its vega risk weight is $55\% \times \sqrt{2} = 77.8\%$ and the vega charge is $\$702{,}488 \times 55\% \times \sqrt{2} = \$546{,}407$.

Component	Charge	Why
Delta	≈ $0	removed by the hedge
Vega	$546,407	the hedge does not touch it
Curvature	$2,239,369	short gamma, both directions
Total	**$2,785,776**	**27.9% of notional**

Delta-hedging removed the delta charge and none of the rest. A book that is flat by the measure a trader watches minute to minute attracts capital equal to more than a quarter of its notional, and 80% of that comes from the curvature term alone.

This is the framework doing deliberately what Chapter 18 says is true. Delta neutrality is a statement about the first derivative at a point; the risk in a short option is the second derivative and the volatility, and a capital rule that let delta-hedging extinguish the charge would be paying for a hedge that does not hedge the exposure.

It also explains a pricing fact from Chapter 35. An issuer selling optionality to retail carries a charge of this kind for as long as it keeps the risk, on a delta-hedged book, so the capital cost is a running cost rather than a one-off, and Section 35.10's fee arithmetic has to cover it before anything is left over.

39.3.3 The default risk charge and the residual risk add-on

The default risk charge treats default as an event the sensitivities cannot see: a bond's spread sensitivity says what a widening costs, not what a default costs. Each position's jump-to-default loss is its loss given default on the notional, less any loss its mark already shows. Longs and shorts in the same issuer offset, offsets between different issuers in the same bucket are recognized only in part, and the net amount is multiplied by a risk weight set by credit quality. A desk long a bond and holding bought protection on the same issuer pays little; a desk long one issuer and short an equal amount of another of the same rating in the same bucket still pays half the long position's charge. The charge is calibrated as the banking book's credit capital is, to a one-year horizon at 99.9%, Section 24.14's worst-case default rate, so that a bond's default capital is broadly comparable whichever book holds it; a bank on internal models computes it by simulating defaults over that year with at least two systematic factors. Migration is left out on purpose, because the spread sensitivities already measure what a downgrade costs.

The residual risk add-on is the framework's admission that some payoffs are beyond its sensitivities. It is 1% of the notional of instruments with an exotic underlying—weather, longevity, future realized volatility (a variance swap's underlying)—and 0.1% of the notional of other instruments bearing residual risks, such as gap risk, correlation risk or behavioral risk (a borrower's option to prepay, for instance). A barrier or a correlation product attracts the add-on because the framework declines to model what its sensitivities miss, which is more honest than a false precision would be.

39.4 THE INTERNAL MODELS APPROACH

A bank with approval computes expected shortfall at 97.5%, Chapter 38's measure at the confidence level Example 38.1 explained, calibrated to a period of stress rather than to recent data. The resulting charge, the **internally modeled capital charge** (IMCC), weights equally the expected shortfall of the whole book, with diversification across risk classes, and the sum

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of the expected shortfalls of each risk class computed alone:

$$\text{IMCC} = \frac{1}{2}\,\text{ES}_{\text{all}} + \frac{1}{2}\sum_{c}\text{ES}_{c}.$$

Here ES_{all} shocks every risk factor together and ES_c only those of risk class c, so the bank's own correlations across risk classes enter with half weight and those within each class in full.

Two design choices follow from Chapter 38's failures. Stress calibration answers Section 38.18's procyclicality directly: the measure cannot fall because markets are calm. And the horizon is no longer a constant.

Example 39.3 The horizon schedule is a capital multiplier

Risk factors are assigned **liquidity horizons** of 10, 20, 40, 60, or 120 days by class:

Horizon	Examples
10 days	major-currency interest rates, large-cap equity prices, major exchange rates
20 days	other interest rates, investment-grade sovereign spreads, small-cap equity prices, other exchange rates, large-cap equity volatility, energy and metal prices
40 days	investment-grade corporate spreads, other sovereign spreads, exchange rate volatility
60 days	high-yield corporate spreads, interest rate volatility, other equity volatility, energy and metal volatility, other commodity prices
120 days	other credit spreads, credit spread volatility, other commodity volatility

Each expected shortfall in the IMCC is computed five times at a ten-day horizon: ES_1 shocks every factor, and ES_j shocks only the factors whose horizon is at least LH_j, the jth horizon in the schedule. The results are aggregated as

$$\text{ES} = \sqrt{\text{ES}_1^2 + \sum_{j\geq 2}\left[\text{ES}_j\sqrt{\frac{\text{LH}_j - \text{LH}_{j-1}}{10}}\right]^2},$$

with weights $\sqrt{(\text{LH}_j - \text{LH}_{j-1})/10}$ of 1.0000, 1.0000, 1.4142, 1.4142 and 2.4495 (taking $\text{LH}_0 = 0$, so that ES_1's weight is one). A factor with a 120-day horizon appears in all five terms, so its contribution is scaled by $\sqrt{1+1+2+2+6} = 3.46$ in total.

Take a desk whose ES values, for the factors surviving at each horizon, run 100, 90, 70, 45, 20. Then

$$\text{ES} = \sqrt{100^2 + 90^2 + (70\times 1.4142)^2 + (45\times 1.4142)^2 + (20\times 2.4495)^2} = 185.34$$

against a flat ten-day figure of 100. The desk's capital is 1.85 times the single-horizon figure for identical positions.

The weights are the square-root-of-time rule of Section 38.14, applied to the stretch of horizon each step adds. Section 38.14 showed that rule failing when returns are autocorrelated, and the framework uses it anyway, because it needs a formula a supervisor can prescribe. The horizon belongs to what is held, not to the rule, so the framework replaces the old single ten-day horizon with five, assigned by risk-factor class, though it keeps that horizon's square-root scaling, and the price is that a bank's capital now depends on a regulatory classification of its risk factors rather than on anything it measures. *The judgment moved rather than disappeared*, which is the recurring shape of this chapter.

39.5 NON-MODELLABLE RISK FACTORS

A factor may only be modeled if there is enough evidence of real prices for it. The **risk factor eligibility test** asks for at least 24 real price observations over the past twelve months, with no 90-day period containing fewer than four, or at least 100 observations over the twelve months. Anything failing that test is a **non-modellable risk factor** (NMRF) and is capitalized by a

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stressed expected shortfall (SES): the loss when that factor alone receives its own stress shock, at least as severe as a 97.5% expected shortfall measured on that factor's own worst stress period, over its liquidity horizon or 20 days if longer.

The penalty is not only that the stress is severe. The non-modellable capital is also added to the modeled charge rather than diversified with it: the modellable part gets the benefit of Chapter 38's covariance structure, and the non-modellable part does not. Within the stress add-on itself, idiosyncratic credit and equity factors are combined as if independent and the rest by a formula with a parameter of 0.6 that amounts to a correlation of 0.36 between any two of them, so some diversification survives there, but none survives between the two parts.

Example 39.4 Two penalties for one illiquid factor

A desk's risk splits into a modellable part with a stand-alone expected shortfall of 80 and an illiquid factor with a stand-alone expected shortfall of 30, correlated at 0.25. If every factor were modellable, the desk would hold

$$\sqrt{80^2 + 30^2 + 2 \times 0.25 \times 80 \times 30} = 92.2.$$

The illiquid factor fails the eligibility test. Its stress scenario, calibrated over its own longer horizon, is assumed here to be 1.6 times its expected shortfall, so its add-on is $1.6 \times 30 = 48$, and the desk holds $80 + 48 = 128$, 38.8% more for identical positions.

The 35.8 difference has two sources of almost equal size. Adding rather than diversifying costs $80 + 30 - 92.2 = 17.8$, and the harsher stress calibration costs $48 - 30 = 18.0$. Only the first is a property of the rule; the 1.6 is this example's assumption.

So identical risk costs more when it sits in an illiquid factor, and that is deliberate: it prices illiquidity, which Section 38.19 put first among what the measure cannot see. The framework cannot model liquidity, so it taxes illiquidity instead—a substitution worth noticing, because the tax falls on the bespoke structured business of Chapter 35.

39.6 THE P&L ATTRIBUTION TEST

The attribution test decides whether a desk keeps its model, and it is the most quantitatively interesting thing in the framework. Approval is granted desk by desk, and a **trading desk** here is a unit the bank designates and the supervisor accepts: a group of traders or accounts running a well-defined business strategy, with its own risk management and a single head.

Each desk produces two daily numbers. **Hypothetical P&L** is the day's profit on yesterday's positions, computed by full front-office revaluation. **Risk-theoretical P&L** is the same thing computed using only the risk factors the risk model contains. If the risk model captures the desk, the two series agree.

The test compares them over 250 days with two statistics. The Spearman rank correlation asks whether the two series rise and fall on the same days. The **Kolmogorov–Smirnov** (KS) statistic is the largest gap between the two series' empirical distribution functions, and asks whether they have the same distribution. A desk is green if the rank correlation is above 0.80 and KS below 0.09, red if the rank correlation is below 0.70 or KS above 0.12, and amber otherwise.

Example 39.5 A third of the P&L may be unexplained

Let a fraction f of the desk's P&L variance come from factors the risk model does not contain, independent of the rest. The risk model revalues the positions with only the factors it contains, so its risk-theoretical series is the hypothetical series without that part: it has variance $1 - f$ of the hypothetical one, and the two correlate at $\rho = \sqrt{1 - f}$. For jointly normal series the rank

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correlation is

$$\rho_S = \frac{6}{\pi}\arcsin\left(\frac{\rho}{2}\right).$$

KS has no closed form here, so the table gives its mean over 2,000 simulated 250-day samples:

Unexplained variance	ρ	Spearman	KS (mean)	Zone
5%	0.9747	0.9722	0.043	green
20%	0.8944	0.8855	0.065	green
30%	0.8367	0.8243	0.081	green
33.8%	0.8135	0.8000	0.088	**amber begins**
40%	0.7746	0.7595	0.099	amber
48.6%	0.7167	0.7000	0.117	**red begins**
50%	0.7071	0.6902	0.120	red

A desk can leave a third of its profit-and-loss variance unexplained by its own risk model and remain in the green zone. It takes half before the model is disqualified.

The two statistics leave green within a couple of percentage points of each other (the rank correlation at 33.8%, the mean KS at about 35%) and reach red together near half, because a missing factor does two things at once. It weakens the day-by-day correlation, which the rank correlation sees, and it shrinks the risk-theoretical series' variance, which KS sees. A model error that did only one of the two would be caught by only one statistic.

The KS column is a mean, and a single desk's KS is noisy. At 30% unexplained variance, where the mean is 0.081, one 250-day sample in four exceeds 0.09, so an otherwise green desk can be pushed into amber by the sample alone.

The practical reading is not that the test is useless. It catches a desk whose risk model misses a factor carrying half its variance, which is what it was built for. What it does not catch is a smaller omission: a missing factor carrying a quarter of the variance, or a model that tracks the hypothetical P&L closely while sitting slightly off it every day.

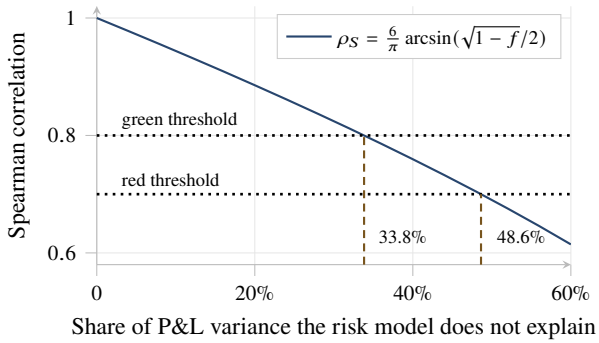


Figure 39.3: Rank correlation against unexplained P&L variance, with the zone thresholds.

Figure 39.3 is worth looking at before designing a risk model. The curve falls steadily, but the green threshold sits low on it: the rank correlation does not fall to 0.80 until a third of the variance is unexplained. The test is permissive because of where its threshold is set, and the choice of statistic adds a second weakness: rank correlation was chosen because it is robust to outliers, and robustness to outliers is robustness to the days that matter.

39.7 BACKTESTING AT DESK LEVEL

The attribution test is not the only hurdle, and the two interact. Each desk also backtests: Chapter 38's exception counting, run at both 97.5% and 99% on the desk's own one-day VaR.

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A desk with more than 12 exceptions at 99%, or more than 30 at 97.5%, over the past 250 days (2.5 and 6.25 on average for a correct model) loses its model approval and moves to the standardized approach.

VaR is backtested even though capital is held against expected shortfall, for the reason Section 38.16 gave: exceptions can be counted, and expected shortfall offers no event to count. The framework checks the quantile and trusts the tail to come from the same model.

Separately, the bank's model as a whole is backtested at 99% across all its approved desks, and that count sets the multiplier of Section 39.9. The desk-level test decides whether a desk may use its model; the bank-level test decides how much the modeled number is scaled up.

The two desk tests catch different faults, which is the point of having both. Backtesting asks whether the measure is *calibrated*: are losses exceeding it at the stated rate? Attribution asks whether the model *contains the right factors*: a factor the model lacks moves the hypothetical P&L but not the risk-theoretical P&L, so the two daily series stop tracking. A desk can pass either and fail the other.

A desk with a correctly sized but structurally wrong model passes backtesting and fails attribution once the missing factor is large. Its VaR is the right magnitude because its sensitivities to the factors it has were scaled up to the desk's total P&L variance, so the risk-theoretical series has the right variance and KS sees little. But the daily series do not track, and the rank correlation alone takes the desk to amber beyond about a third of the variance and to red near half (Example 39.5). A desk with the right factors and a stale calibration does the reverse: hypothetical and risk-theoretical P&L move together, so attribution passes, while exceptions pile up because the level is wrong.

The combination is more powerful than either, and it is the closest the framework comes to actually validating a model. It is still weak in the sense Section 38.16 established (250 days, and Example 39.5's permissiveness), but the two tests fail on different errors, so a model that survives both is either wrong only moderately or wrong in what neither measures, such as the shape of the tail beyond the quantiles 250 days can test.

39.8 DESK-LEVEL APPROVAL, AND THE CLIFF

Approval is granted and withdrawn *desk by desk*, which is the framework's most important structural choice. Figure 39.4 sets out the sequence of tests a desk passes through.

A failing desk does not disqualify the bank, so the consequence is proportionate. But a red desk, or one that fails its backtest, moves to the standardized charge, which for a book with genuine diversification can run well above the model number. An amber desk keeps its model but pays a surcharge: half the excess of the standardized over the modeled charge for the green and amber desks together, scaled by the amber desks' share of the standardized charge. The total is in turn capped at what the standardized approach would require for every desk, except that any excess of the approved desks' modeled charge over their own standardized charge is still held, so the only saving the cap can deliver is the diversification the standardized calculation itself recognizes when all desks are computed together.

The move to red is a cliff, not a slope, and it is triggered by a statistical test on 250 days, which Section 38.16 already showed to be a short sample. So a desk's capital can rise by two-thirds (from 60 to 100 for the desk of Example 39.7 below) on evidence that would not settle a scientific question, and the rational response is to run risk models that are conservative in the specific sense of containing every factor the front office prices with, whether or not they matter. The amber surcharge softens the first step down without removing the cliff at the bottom.

That incentive is the framework working as intended. It cannot verify a model, so it makes the cost of divergence high enough that banks align their risk models with front-office pricing

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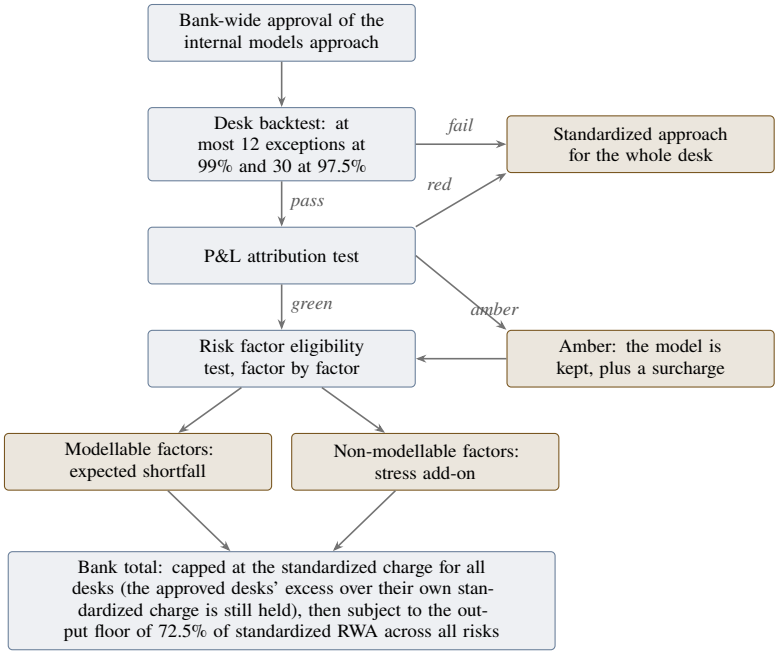


Figure 39.4: The tests a trading desk passes through, and where each outcome leads.

themselves.

39.9 ASSEMBLING THE TOTAL

The pieces above combine in a way worth seeing once, because the arithmetic explains where a trading book's capital actually comes from. The modeled part is

$$\max\Big(\mathrm{IMCC}_{t-1} + \mathrm{SES}_{t-1},\ \ m_c \times \overline{\mathrm{IMCC}} + \overline{\mathrm{SES}}\Big),$$

where IMCC is the internally modeled capital charge of Section 39.4, SES the non-modellable stress add-on, $t-1$ marks the previous day's figure, the bars denote sixty-day averages, and $m_c \geq 1.5$ is a multiplier that rises with the bank-wide backtesting exceptions of Section 39.7.

Example 39.6 Where the capital comes from

Take a bank with $\mathrm{IMCC}_{t-1} = 420$, $\overline{\mathrm{IMCC}} = 380$, $\mathrm{SES}_{t-1} = 95$ and $\overline{\mathrm{SES}} = 88$:

$$\max(420 + 95,\ \ 1.5 \times 380 + 88) = \max(515,\ 658) = 658.$$

The averaged, multiplied term binds, which is normal: with a multiplier of at least 1.5 it starts half again above a typical day's charge, so the requirement depends on a sixty-day average rather than on yesterday. Only a sharp rise in the latest figure reverses that.

The multiplier is where backtesting enters. The bank-wide exception count over 250 days sets m_c, with the zone boundaries of Section 38.16's traffic light, whose yellow zone the current framework calls amber:

Multiplier	Modeled charge	Increase
1.50 (green)	658	—
1.70 (amber, five exceptions)	734	+11.6%
2.00 (red)	848	+28.9%

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Adding the default risk charge on the approved desks and the standardized charge on the desks with no model approval gives the total. The residual risk add-on belongs to the standardized charge, so it sits inside the standardized-approach line:

Component	Amount	Share
Modeled measure (ES and NMRF)	658	60.2%
Default risk charge, approved desks	140	12.8%
Desks on the standardized approach	295	27.0%
of which residual risk add-on	35	3.2%
Total	**1,093**	100%

More than a quarter of the requirement comes from desks whose models were not approved, and that share is the framework's real lever. Hedging moves this number at the margin; passing the attribution test on more desks moves it wholesale—which is why the test of Section 39.6, and not the measure of Section 39.4, is what trading-book capital management is actually about.

39.10 THE OUTPUT FLOOR

One provision sits above everything in this chapter and can nullify all of it. The **output floor** caps the benefit of internal models across *all* risk types together: a bank's total RWA may not fall below 72.5% of what the standardized approaches would produce, once the floor is fully phased in: by 2028 under the Basel timetable, by 2030 in the European Union and the United Kingdom, and in China from 2024, where it replaced an earlier 80% floor. Canada has held its floor at 67.5% since 2025. It is not a market risk rule, which is exactly why it matters here: the trading book's modeled capital is one input to a constraint that binds at the level of the whole bank.

Example 39.7 The same desk failure, three banks, three costs

Each bank has standardized RWA of 100 for trading and 900 for everything else, so its floor is $0.725 \times 1{,}000 = 725$. Its trading book is a single desk, whose RWA are 60 on models and 100 on the standardized approach.

	Modeled RWA	Floor	Held
Bank A — modeled non-trading RWA 700			
Desk keeps approval	760	725	760
Desk loses approval	800	725	800
Cost of failing			**40**
Bank B — modeled non-trading RWA 600			
Desk keeps approval	660	725	**725**
Desk loses approval	700	725	**725**
Cost of failing			**0**
Bank C — modeled non-trading RWA 640			
Desk keeps approval	700	725	**725**
Desk loses approval	740	725	740
Cost of failing			**15**

The identical desk, failing the identical test, costs one bank 40, another 15 and the third nothing. For Bank B the floor already binds whatever the desk does, so the trading book's modeled number is slack and improving it changes no requirement at all. Bank C sits between: the floor absorbs 25 of the desk's 40, and only the part that lifts the bank above the floor is a cost.

This inverts the incentive the rest of the chapter carefully constructs. Sections 39.6 and 39.8 argued that the attribution test disciplines banks because failing it is expensive. For a floored

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bank failing it costs nothing, and every other refinement earns nothing: a better model, a narrower NMRF population, a tighter liquidity classification. The tests still run, and they no longer change the requirement.

This explains a pattern that otherwise looks irrational. Section 39.13 notes that banks have chosen standardized capital over model approval on some desks. For a bank near the floor that is simply correct: model approval carries real operational cost and, below the floor, delivers no capital benefit whatever. For such a bank the framework's preferred approach is unattractive not because the modeling is hard but because a different rule has already decided the answer.

Whether this is a defect depends on what the floor is for. It exists because supervisors do not believe model outputs are comparable across banks: the same distrust that produced every test in this chapter, applied at the top rather than desk by desk. Having both is redundant by construction, and the redundancy is deliberate: the tests police the model, and the floor limits the damage if the tests fail.

39.10.1 Where the modeled credit number comes from

Example 39.7 took each bank's modeled non-trading RWA as given. Most of it is credit risk in the banking book, and for a bank approved to use the **internal ratings-based approach** (IRB) it comes from a formula that is Section 24.14's worst-case default rate with the supervisor's parameters in it. The bank supplies each exposure's probability of default (PD), its loss given default (LGD) and its effective maturity M; under the foundation version it supplies only the PD, and the LGD of a senior unsecured claim on a nonfinancial corporate is set at 40%. The capital per dollar of exposure is

$$K = \text{LGD}\left[N\left(\frac{N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(0.999)}{\sqrt{1-\rho}}\right) - \text{PD}\right]\frac{1 + (M-2.5)\,b}{1 - 1.5\,b} \tag{39.1}$$

and the risk weight is $12.5K$, where 12.5 is the reciprocal of the 8% minimum capital ratio; ρ is a prescribed correlation and b a maturity coefficient, both given below.

Three choices in (39.1) are the supervisor's rather than the bank's. The confidence is 99.9% over one year. The correlation (the R of the Basel text, written ρ here as in Section 24.14) is prescribed rather than estimated, falling from 0.24 for the safest corporate borrowers to 0.12 for the riskiest,

$$\rho = 0.12\,w + 0.24\,(1-w), \qquad w = \frac{1 - e^{-50\,\text{PD}}}{1 - e^{-50}}$$

on the view that a strong firm defaults mostly when the economy does and a weak one mostly for reasons of its own. The last factor is a **maturity adjustment**, with $b = (0.11852 - 0.05478 \ln \text{PD})^2$, which equals one at a one-year maturity and charges a longer loan for the downgrade risk that Example 24.10 measured. The PD itself is floored at 0.05%.

The formula can be applied one loan at a time because, in a one-factor model of an infinitely granular portfolio, the portfolio's quantile is the sum of the loans' own worst-case losses. That property is what lets a supervisor write a capital rule at all, and it is also the formula's blind spot: a book concentrated on a few large borrowers or exposed to a second systematic factor, a region or an industry, holds more risk than the sum says. Neither is charged in the formula; both are left to the supervisor's own review of each bank, outside the minimum requirement.

Example 39.8 Where the floor bites on a loan book

Take a corporate loan at each rating, with the one-year default probability from the last column of Table 24.2, an LGD of 40% and a maturity of 2.5 years, and compare its IRB risk weight with the standardized risk weight for a rated corporate in Table 27.1.

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Rating	PD	ρ	WCDR	Maturity adj.	IRB RW	Std. RW	Ratio
A	0.05%	0.237	2.04%	1.752	17.5%	50%	0.35
BBB	0.28%	0.224	6.89%	1.411	46.7%	75%	0.62
BB	1.52%	0.176	16.95%	1.222	94.2%	100%	0.94
B	6.45%	0.125	32.42%	1.121	145.6%	150%	0.97
CCC	30.60%	0.120	72.59%	1.053	221.1%	150%	1.47

The BBB row runs as follows:

$$w = \frac{1 - e^{-50\times 0.0028}}{1 - e^{-50}} = 0.1306, \qquad \rho = 0.12 \times 0.1306 + 0.24 \times 0.8694 = 0.2243,$$

$$\text{WCDR} = N\left(\frac{N^{-1}(0.0028) + \sqrt{0.2243}\, N^{-1}(0.999)}{\sqrt{1 - 0.2243}}\right) = N(-1.4837) = 6.895\%,$$

$$b = (0.11852 - 0.05478 \ln 0.0028)^2 = 0.1941,$$

$$K = 0.40 \times (0.06895 - 0.0028) \times \frac{1 + (2.5 - 2.5) \times 0.1941}{1 - 1.5 \times 0.1941} = 3.733\%,$$

and the risk weight is $12.5K = 46.7\%$.

The ratio column is the one that matters for the floor. A loan rated A or BBB is modeled at well under 72.5% of its standardized weight (35% of it at A, 62% at BBB), so a bank whose book is mostly A and BBB lending, the typical book of a large bank lending to large companies, sits on the floor whatever its models say. Speculative-grade loans are modeled at close to or above their standardized weight, well above 72.5% of it, and leave the floor slack. Bank B of Example 39.7 is therefore not a curiosity. It is what a bank with a high-quality loan book looks like, and for such a bank the trading desk's model approval is worth nothing because its lending has already put it on the floor.

The standardized side matters too. The United States does not use credit ratings in its current capital rules, and its standardized approach puts most corporate exposures at 100%, against which even the BBB loan is modeled at under half, so the floor bites harder there. The European Union went the other way and, until 2032, lets a bank's floor calculation weight an unrated corporate exposure at 65% where its own PD estimate is no more than 0.5%, precisely to soften the floor on investment-grade lending.

Figure 39.5 draws the whole curve. The IRB curve runs below 72.5% of the standardized weight at both the A and BBB points of the Basel table, so that is where a loan book pushes a bank onto the floor.

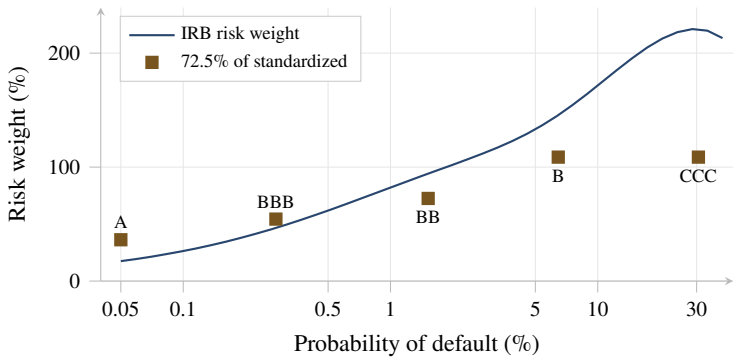


Figure 39.5: IRB risk weight for a corporate loan (LGD 40%, maturity 2.5 years) against 72.5% of the standardized weight at each rating's default probability.

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39.11 PRUDENT VALUATION

There is a second capital channel that has nothing to do with the measures above, and for an exotic book it can matter more. In the European Union and the United Kingdom it is a rule; elsewhere supervisors expect valuation reserves of the same kind without prescribing them.

Fair value is an exit price, typically marked at mid within the bid–ask spread or at a model price. Prudent value is a price the position could actually be realized at with 90% confidence, and the difference, the **additional valuation adjustment** (AVA), is deducted from common equity directly rather than risk-weighted.

Example 39.9 The deduction on a $100 million book

An exotic book with a fair value of $100 million carries the following adjustments. Three of them, market price uncertainty, close-out cost and model risk, overlap in what they measure, so the core approach of the European rules, as against the simplified approach for small books, adds them and then counts half of the sum. The rules split the unearned-credit-spread and funding adjustments into the same three parts, so they join that sum.

Component	Basis points	Amount
Market price uncertainty	40.0	$400,000
Close-out cost (bid–ask)	35.0	$350,000
Model risk	60.0	$600,000
Unearned credit spread	10.0	$100,000
Investing and funding costs	8.0	$80,000
Sum, counted at 50%	76.5	$765,000
Concentrated position	25.0	$250,000
Future administrative costs	6.0	$60,000
Early termination	5.0	$50,000
Operational risk	4.0	$40,000
Total	**116.5**	**$1,165,000**

Model risk is the largest single component in this book, and it is the one a desk cannot reduce by trading better: a tighter market narrows the close-out cost and does nothing to the model uncertainty of a payoff nothing quotes.

The capital arithmetic makes the size clear. A $1.165m deduction from common equity, at an assumed total requirement of 12% of RWA, removes the capital that would support $9.7m of risk-weighted assets, equity the bank can no longer use for anything else, whatever the position's own risk weight.

This is where Section 21.11's model spread becomes a number on a balance sheet. Two models agreeing on every vanilla and disagreeing on an exotic produce a valuation uncertainty; that uncertainty is not a modeling curiosity but the input to a deduction, and it is why dealers maintain model reserves against exotic books instead of relying on a single model's mark.

The measurement that makes this possible comes from Hull and Suo (2002), and the regulatory framing arrived afterward. Their procedure (calibrate competing models to a common surface, verify each reproduces it, then price what the surface does not determine) turns model risk from a caveat into a quantity, and a quantity is the only thing a capital rule can act on. The prudent valuation regime is that idea written as a requirement: it does not ask whether a model is right, which no rule could adjudicate, but how far defensible alternatives disagree.

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39.12 STRESS TESTING AND WHAT SITS OUTSIDE THE FORMULA

Everything above is a formula, and supervisors do not rely on formulas alone.

Supervisory stress tests apply a common scenario across banks and compare. Their value is the comparison rather than the scenario: a bank whose losses are unusual under a shared shock is revealing a concentrated position, and the exercise finds it without anyone specifying what to look for. In the United States the exercise also sets capital: the largest banks' trading books take an instantaneous global market shock, and the losses feed the **stress capital buffer**, a requirement on top of the minimum that rises with the bank's projected stress losses. For those banks a supervisory scenario, not the formula above, often decides how much the trading book costs.

Reverse stress testing inverts the question—not "what do we lose in this scenario" but "what scenario makes us fail"—and it is the only exercise in the framework that begins from the outcome. It is also the one most likely to find Section 38.19's blind spots, because it is not constrained to the risk factors the model contains. Section 38.17 works one through for a two-asset book, and finds that the most plausible route to a given loss runs through the position carrying the largest share of the VaR.

And the supervisory review sets capital above the formula where the formula is known to be inadequate; concentration, which Section 38.19 said no distribution-based measure sees, is the standard case.

The pattern across all three is the same. The formula handles what can be specified in advance; the discretionary machinery handles the rest, and the rest is where Chapter 40's losses live.

39.13 WHAT THE FRAMEWORK CHANGED, AND WHO BEARS THE COST

Three things changed, and only the first was intended. Capital went up for trading, and unevenly. The increase falls hardest on exotic and illiquid books (through the residual risk add-on, the NMRF treatment, and the long liquidity horizons) and lightly on high-volume trading in liquid instruments.

The internal models approach became less attractive than it was designed to be. Between the operational cost of the attribution test, the NMRF data requirements, and the cliff risk, many banks concluded that standardized capital on some desks was cheaper than model approval (the PRA reported in June 2026 that few UK firms planned to adopt the internal models approach), which is the opposite of the intended direction, and Section 39.10 showed why the floor makes that conclusion correct for some of them. A framework that makes its own preferred approach unattractive has a design problem.

The business consolidated. Fixed costs favor scale: the data infrastructure for modellability, the daily attribution machinery, and the desk-level governance are nearly independent of volume. Chapter 35's structured issuance and Chapter 27's client clearing both concentrated for this reason, and the trading book is a third instance. The cost lands in four places.

Dealers bear it directly and pass what they can into spreads, which is why the pricing of Chapter 35's structured issuance and Chapter 34's exotics carries a capital component that the models in those chapters do not contain.

Clients bear it as a wider bid–ask on precisely the trades the framework taxes (bespoke, illiquid, long-dated), and the effect is to push demand toward standardized products, which is intended.

Smaller dealers bear it as a fixed cost and some exit, so the framework's second-order effect

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is concentration, and Section 27.12's concern about concentration is partly a consequence of the rules in this chapter.

And supervisors bear the residual: a framework that cannot verify models must set incentives instead, so the quality of the outcome depends on the calibration of penalties rather than on the correctness of any measurement.

39.14 MODEL RISK

Everything above assumes the model maps market factors to portfolio value correctly, and that assumption is itself an exposure. *Model risk is not a subset of market risk*, which is why it needs its own treatment. A market risk measure is conditional on the pricing model: the model that marks the position also supplies the revaluations its risk is measured from, and the attribution test compares the risk model with that same pricer, never the pricer with anything outside it. An error in the pricer misstates the mark and the risk at once, and nothing inside the system disagrees with it.

Section 21.11 made this concrete. Two models calibrated to identical vanilla prices give different values for the same exotic, and the difference is not measurement error—it is the part of the payoff the calibration instruments do not span. No amount of data resolves it, because the disagreement is about what happens where nothing trades.

The governance response is well established and modest in its claims. Models are inventoried; each has an owner and a stated purpose; validation is performed independently of development; limitations are documented and the documented limitation is the basis for a reserve or a capital add-on. In the United States the supervisory guidance is SR 26-2 (2026), which replaced the 2011 guidance known as SR 11-7 and scales the effort to the risk a model creates; in the United Kingdom it is the PRA's SS1/23, in force since May 2024. Neither document verifies a model, and neither claims to: they make the assumptions explicit and assign someone to own them.

39.15 VALIDATING WHAT CANNOT BE VERIFIED †

A model is *verified* when its output can be checked against something independent: an analytic formula, a converged numerical method, a market price. Validation is the weaker activity of testing a model that cannot be checked that way, and two kinds of model force the distinction.

The first is the calibrated model of Section 21.11. Local and stochastic volatility models fitted to the same vanilla surface agree on every vanilla and disagree on a barrier, and nothing that trades can say which is right. Validation can only measure the disagreement, which is what Hull and Suo's procedure and Section 39.11's model risk adjustment do, and cannot remove it.

The second is the learned model. A **learned pricer** is a neural network trained to reproduce a pricing model's outputs, so that a slow calculation can be evaluated in microseconds; a **learned hedging policy** is a network trained on simulated price paths to choose positions that minimize a risk measure such as Chapter 38's expected shortfall. The online note [TN41] treats both, and three of its findings bear on validation.

The training set is part of the model. A learned pricer degrades silently outside the range of inputs it was trained on, returning a plausible number rather than an error ([TN41], Section 3). A model inventory that records the network and omits the training range has recorded the less important half, and retraining on new data is a model change requiring revalidation, not a refresh.

Reproducibility is not the same as determinism. In Section 12 of [TN41], three training runs of one hedging policy differed only in the random number seed, agreed on the expected shortfall they achieved to within 0.8%, and held positions that differed by up to 1.7 percentage

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points of delta. A validation standard that asks whether results reproduce will pass all three, although the positions they would actually hold differ.

Validation reduces to out-of-sample testing in the region someone happened to sample. A Black–Scholes implementation can be checked against an analytic formula; a learned pricer can only be checked against the pricer it learned from, and the case study in Section 4 of [TN41] is the case where that pricer was wrong and the network smoothed the evidence.

The existing framework is adequate in structure and strained in practice. Inventory, ownership, independent validation, and documented limitations are the right categories, and SR 26-2 applies them to non-generative machine-learning models such as these, leaving only generative and agentic artificial intelligence (AI) outside its scope. What is missing is a validation technique with the power to distinguish a good model of either kind from a plausible one, and until there is one, the sensible response is the conservative one: use such models where a classical answer exists to check them against.

SUMMARY

A capital rule is a contract about a number the bank itself produces, so the framework both specifies the model and tests it. Each complication answers a specific failure: the prescriptive boundary removes an arbitrage, stress calibration answers procyclicality, and liquidity horizons make the horizon depend on what is held, though only by risk-factor class. A desk whose horizons are spread across the schedule pays materially more than under a flat ten-day measure with no change in any position.

The standardized charge refuses to trust a correlation. It takes the worst of the prescribed correlations, a higher set and a lower set, and which scenario binds is a property of the book: a directional position is worst under high correlation and a butterfly under low, by a far wider margin. Curvature follows Chapter 18's economics: long optionality attracts no charge, and the same option sold attracts all of it. Non-modellable factors are capitalized by a stress scenario added without diversification, so a book with a sizable non-modellable share pays markedly more capital for identical risk.

The P&L attribution test is more permissive than its reputation. A desk may leave a sizable share of its P&L variance unexplained and stay green, because both statistics leave the green zone only once a missing factor carries roughly a third of the variance, so the test catches a large missing factor but not a moderate one. Desk-level backtesting checks the model's calibration and attribution its content, and a model that passes both is wrong only moderately or in what neither sees. Approval is withdrawn desk by desk against a 250-day sample, so capital moves as a cliff, softened only by the amber surcharge.

The output floor holds risk-weighted assets at no less than 72.5% of the standardized result when fully phased in, so for a bank where the floor already binds, failing the attribution test costs nothing, which is why declining model approval can be correct. Most of the modeled number the floor compares is banking-book credit, set by the IRB formula, which is the worst-case default rate at 99.9% with a prescribed correlation and a maturity adjustment; it models A and BBB lending at well under 72.5% of the standardized weight, so the floor binds hardest on banks with the best loan books. Prudent valuation deducts from equity directly, and for an exotic book model risk can be its largest component. Supervisory stress tests and review set capital where the formula cannot, and in the United States a stress scenario often decides the trading book's cost.

Model risk is not a subset of market risk: the pricing model that marks a position also feeds its risk measure, so an error in it misstates both at once, and no test in the framework compares it with anything outside itself. Governance makes assumptions explicit without verifying anything, and models that cannot be verified (calibrated models that disagree where

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nothing trades, and learned models) strain it most, so the conservative use is the defensible one.

FURTHER READING

The Basel Committee's *An Explanatory Note on the Basel II IRB Risk Weight Functions* (2005) derives the formula of Section 39.10 and explains each of its parameter choices, and Gordy (2003) proves the portfolio invariance that lets it be applied one loan at a time. The Basel Committee's *Minimum Capital Requirements for Market Risk* (2019) is the primary source for everything from Section 39.2 to Section 39.9; the sections motivating the boundary and the liquidity horizons are the ones to read first, and the industry quantitative impact studies are where the capital effects of Section 39.13 are actually measured. The output floor is in the Committee's *Basel III: Finalising Post-Crisis Reforms* (2017), which is the document to read for how the floor was meant to interact with internal models. On prudent valuation, the European Banking Authority's regulatory technical standards, adopted as Commission Delegated Regulation (EU) 2016/101, set out the nine adjustments and their aggregation. Jurisdictional implementation differs in ways that matter and in ways that do not; the divergences worth tracking are the treatment of internal models and the boundary, and both are documented in supervisory consultation papers rather than in the standard. On model risk, the US interagency guidance SR 26-2 (2026) and the PRA's SS1/23 (2023) are the current frameworks, and the earlier SR 11-7 (2011) is worth reading for the vocabulary both inherited and because it makes no claim to verify anything. Hull and Suo (2002) is the paper behind Section 39.11's measurement of model risk. On the questions raised in Section 39.15, SR 26-2's footnote on generative and agentic AI models is where US supervisors currently draw the line.

PRACTICE QUESTIONS

39.1 Explain why the trading-book boundary was made prescriptive, compute the capital at stake in Section 39.2's bond portfolio, and explain why removing the capital benefit of reassignment matters more than tightening the definition.

39.2 Explain why the standardized approach supplies correlations rather than estimating them, and relate this to Section 38.11.

39.3 Using the weighted sensitivities and correlations of Example 39.1, compute the directional book's bucket charge under the high-correlation scenario, and explain why it equals the sum of the weighted sensitivities.

39.4 Using Example 39.1, explain why the directional book is worst under high correlation and the butterfly under low, why the butterfly's charge falls to zero under high correlation, and why the hedged book nonetheless pays the larger proportional penalty.

39.5 Explain why a long option attracts no curvature charge while the same option sold attracts the whole of it, using Figure 39.2.

39.6 Compute the curvature charge on a short \$10 million at-the-money six-month put on the stock of Section 39.3 at 25% volatility and zero interest rates, and explain why it equals the short call's.

39.7 Recompute Example 39.2 for a small-cap stock, whose volatility has a 60-day liquidity horizon, keeping the curvature shock at 55% so that only the vega term changes, and state the total charge as a share of notional and the curvature term's share of it.

39.8 A desk holds \$50 million notional of variance swaps and \$200 million of products carrying gap risk. Compute its residual risk add-on, and explain why the framework charges a flat percentage rather than modeling these payoffs.

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39.9 In Example 39.3, a factor class is reclassified from a 40-day to a 60-day horizon, raising ES_4 from 45 to 60. Compute the new aggregate, and explain why the desk's capital changed although nothing it holds did.

39.10 State the risk factor eligibility test, and redo Example 39.4 with a correlation of 0.50 between the two parts, separating the cost of lost diversification from the cost of the stress calibration.

39.11 Using the closed form of Example 39.5, compute the share of unexplained variance at which the rank correlation equals 0.75, and state the zone the desk is in.

39.12 Explain why the rank correlation and the Kolmogorov–Smirnov statistic leave the green zone at nearly the same share of unexplained variance, and construct a model error that only one of them would detect.

39.13 Explain what kind of model failure the attribution test catches and what kind it does not, and relate the second to a learned pricing or hedging model.

39.14 A desk records 9 exceptions at 99% and 34 at 97.5% over 250 days. State the consequence, and explain why VaR is backtested when capital is held against expected shortfall.

39.15 Explain what backtesting tests that attribution does not, and construct a desk that passes each while failing the other.

39.16 Explain why desk-level approval is a cliff rather than a slope, what the amber surcharge changes, and what behavior the cliff induces.

39.17 Using Example 39.6, recompute the modeled charge when yesterday's internally modeled charge is 650, state which term binds, and explain why the averaged term usually does.

39.18 State how the bank-wide backtesting exception count enters the capital formula, compute the increase from a green multiplier to the amber multiplier for seven exceptions (1.83), and explain why the increase is smaller than the rise in the multiplier.

39.19 Using Example 39.6, state the share of the total requirement contributed by desks without model approval, and explain why this makes the attribution test the focus of capital management.

39.20 Using Example 39.7, find the modeled non-trading RWA at which failing the desk's test costs exactly 20, and explain what the floor does to the incentives of Section 39.6.

39.21 Explain why a bank near the output floor may rationally decline model approval, and why this is not evidence that its modeling is inadequate.

39.22 Using Example 39.9, compute the deduction if the adjustments in the 50% sum were added in full rather than at 50%, explain why model risk cannot be reduced by trading better, and compute the risk-weighted assets the full deduction would displace at a 12% capital requirement.

39.23 Explain how a supervisory stress test becomes trading-book capital in the United States, and why reverse stress testing is the exercise most likely to find what the formula misses.

39.24 Explain why model risk is not a subset of market risk, and why the error and the measure move together.

39.25 Using Section 39.15, explain why reproducibility is an inadequate validation standard for a learned hedging policy, and what should be required instead.

39.26 A senior unsecured loan to a nonfinancial corporate has a one-year PD of 1% and a maturity of 2.5 years, and the bank uses the foundation IRB approach. Compute the prescribed correlation, the worst-case default rate, the maturity adjustment, the capital

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requirement and the risk weight, and state whether the output floor binds on a book of such loans if the standardized weight is 100%.

39.27 Using Example 39.8, explain why the output floor binds on a bank that lends mainly to investment-grade companies and not on one that lends mainly to speculative-grade companies, and what this implies for the value of trading-book model approval at each.

FURTHER QUESTIONS

39.28 Example 39.5 assumes the unexplained component is independent noise. Redo the analysis with an unexplained component that is systematically related to the explained one (a missing factor correlated with the ones present) and determine whether the test becomes more or less powerful.

39.29 Example 39.5 shows the two statistics reacting to a missing factor at nearly the same point. Construct a desk whose model error makes the Kolmogorov–Smirnov statistic bind well before the rank correlation, and state whether that error is more or less serious than the one the correlation catches.

39.30 Section 39.8 argues the cliff induces banks to include factors that do not matter. Model the bank's optimization explicitly (capital cost against the probability of failing the test) and determine the conditions under which adding an immaterial factor is rational.

39.31 Example 39.3 treats liquidity horizons as a fixed schedule. Estimate empirical close-out horizons for three instrument classes, compare them with the regulatory assignment, and determine the capital effect of the difference.

39.32 Section 39.13 argues that fixed costs concentrate the market. Build a model of dealer entry and exit under a fixed compliance cost, calibrate it to the observed consolidation, and determine what level of fixed cost the observed market structure implies.

39.33 Section 39.15 argues that no current validation technique separates a good learned model from a plausible one. Propose one, state what it would need to observe, and determine whether the required data exists for a pricing surrogate and for a hedging policy respectively.

39.34 The framework taxes illiquidity because it cannot model it. Design a capital treatment that charges for liquidity directly, state what it would require banks to report, and determine whether the reporting burden is smaller or larger than the current non-modellable-factor machinery.

39.35 Section 39.10 shows a floor-bound bank indifferent to its desk tests. Model a bank's choice, desk by desk, between internal models and the standardized approach as a function of its distance from the floor, and determine the distance at which the internal models approach stops paying for its fixed cost.

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